AGENTIC-ARBITER -- FREE-COOLING DECISION REPORT

Data centres over-cool, continuously, because nobody can promise them the hours ahead. A chiller plant needs hours of notice to change mode, and a thermometer only ever reports NOW -- so the mechanical chillers keep running through hours that outside air could have cooled for free. FortyGuard closes that gap with heat intelligence 2 m above the ground, the height a ground-mounted condenser actually breathes. This agent turns that forecast into an hour-by-hour schedule carrying a calibrated safety bound.

THE SITE

Location	IL facility IL_way_1316005112 -- station KORD, America/Chicago
Building	OSM 1316005112 OSM way 1316005112
Facade gap	not applicable -- one building, no second facade
Weather record	43,775 real hourly records from KORD
Plume physics	NOT MODELLED. No other tagged data centre lies inside the solver's validated range, so there is no neighbour intake for a plume to arrive at: the quantity does not exist here rather than being unmeasured. This is a statement about the model's domain, not a claim that recirculation here is zero. A building's own exhaust re-entering its own intake is not modelled at any site in this project.

FortyGuard data: weather and geometry are this site's own; the four measured FortyGuard level offsets and the N-26 coverage record are Ashburn's, because only Ashburn has forecast/outcome day pairs. Quote the hours for this site; quote the coverage for Ashburn.

THIS REPORT IS A SNAPSHOT, NOT THE LIVE PAGE

The interface recomputes the decision for whatever configuration a reader selects. This file was generated at build time for the ONE configuration named below, chosen because it is the configuration in which this site's agent does the most while still changing mode at least once. If the numbers on screen differ from the numbers here, the configuration differs -- compare the two lists before concluding anything else.

Day	2022-05-15 (chatter)
Plant limit	21.0 C
Notice required	3 h
Forecast skill	0.50 relative to persistence
Level anchor	sensor
Condenser bank	longest facade
Switch budget	2 mode changes per day
Minimum dwell	3 h
Max dew point	15.0 C (Green Grid WP#46 p.6)

WHAT THE AGENT DID ON THIS DAY

The agent ran free cooling for 8 of 24 hours with 1 mode change. That is 3 more than the reactive incumbent, at 0 unsafe hours against its 0. 1 hour was safe but still ran chillers, because the schedule could not afford them.

Agent	8 of 24 hours free cooling, 1 mode change(s), 0 unsafe hour(s)
Incumbent	5 of 24 hours free cooling, 1 mode change(s), 0 unsafe hour(s)

1 hour(s) passed every gate and still ran chillers, because the schedule could not afford them

EVERY HOUR, AND THE REASON FOR IT

hh	mode	safe	binding	bound	limit	actual	margin
00	mechanical	no	dry-bulb	21.385	21.0	21.110	0.911
01	mechanical	no	dry-bulb	21.110	21.0	20.560	0.863
02	mechanical	no	dry-bulb	21.110	21.0	20.560	0.825
03	mechanical	no	dry-bulb	21.385	21.0	21.110	0.762

04	mechanical	no	dry-bulb	21.115	21.0	21.110	0.710
05	mechanical	yes	-	20.840	21.0	20.560	0.615
06	mechanical	no	dry-bulb	21.395	21.0	20.560	0.661
07	mechanical	no	dry-bulb	22.775	21.0	21.110	1.225
08	mechanical	no	dry-bulb	23.335	21.0	20.560	1.665
09	mechanical	no	dry-bulb	25.005	21.0	22.780	1.709
10	mechanical	no	dry-bulb	25.555	21.0	23.330	1.594
11	mechanical	no	dry-bulb	25.280	21.0	24.440	1.230
12	mechanical	no	dry-bulb	24.445	21.0	21.110	1.263
13	mechanical	no	dry-bulb	24.165	21.0	21.110	1.055
14	mechanical	no	dry-bulb	24.165	21.0	21.110	0.846
15	mechanical	no	dry-bulb	22.500	21.0	21.670	0.921
16	FREE	yes	-	20.550	21.0	18.330	1.008
17	FREE	yes	-	20.280	21.0	18.890	0.830
18	FREE	yes	-	20.000	21.0	18.330	0.864
19	FREE	yes	-	18.055	21.0	17.780	1.076
20	FREE	yes	-	18.055	21.0	17.220	1.096
21	FREE	yes	-	17.220	21.0	16.110	0.964
22	FREE	yes	-	16.945	21.0	16.110	0.801
23	FREE	yes	-	16.665	21.0	15.560	0.953

bound = the agent's 90 %-nominal upper bound on intake air, forecast plus margin
actual = what the intake temperature turned out to be, from the station record
margin = group-conditional forecast error for that hour of day, plus plume spread

THE REASONING, HOUR BY HOUR

00:00 MECHANICAL [dry-bulb]
Mechanical. The upper bound on intake air is 21.385 C against a 21.0 C limit, so it fails by 0.385 C. It would take a limit 0.385 C higher, or a bound 0.385 C tighter, to change this hour.

01:00 MECHANICAL [dry-bulb]
Mechanical. The upper bound on intake air is 21.110 C against a 21.0 C limit, so it fails by 0.110 C. It would take a limit 0.110 C higher, or a bound 0.110 C tighter, to change this hour.

02:00 MECHANICAL [dry-bulb]
Mechanical. The upper bound on intake air is 21.110 C against a 21.0 C limit, so it fails by 0.110 C. It would take a limit 0.110 C higher, or a bound 0.110 C tighter, to change this hour.

03:00 MECHANICAL [dry-bulb]
Mechanical. The upper bound on intake air is 21.385 C against a 21.0 C limit, so it fails by 0.385 C. It would take a limit 0.385 C higher, or a bound 0.385 C tighter, to change this hour.

04:00 MECHANICAL [dry-bulb]
Mechanical. The upper bound on intake air is 21.115 C against a 21.0 C limit, so it fails by 0.115 C. It would take a limit 0.115 C higher, or a bound 0.115 C tighter, to change this hour.

05:00 MECHANICAL
Mechanical EVEN THOUGH THIS HOUR IS SAFE. Every gate passes -- the bound is 20.840 C against a 21.0 C limit. The schedule is what forbids it: no relaxation of either the switch budget or the dwell would free it, so the surrounding hours are worth more. This is the one explanation a thermostat cannot give, because a thermostat has no plan to be constrained by. It is also why the decision is a schedule and not a comparison: spending a mode change here would cost a better one later.

06:00 MECHANICAL [dry-bulb]
Mechanical. The upper bound on intake air is 21.395 C against a 21.0 C limit, so it fails by 0.395 C. It would take a limit 0.395 C higher, or a bound 0.395 C tighter, to change this hour.

07:00 MECHANICAL [dry-bulb]
Mechanical. The upper bound on intake air is 22.775 C against a 21.0 C limit, so it fails by 1.775 C. It would take a limit 1.775 C higher, or a bound 1.775 C tighter, to change this hour.

08:00 MECHANICAL [dry-bulb]
Mechanical. The upper bound on intake air is 23.335 C against a 21.0 C limit, so it fails by 2.335 C. It would take a limit 2.335 C higher, or a bound 2.335 C tighter, to change this hour.

09:00 MECHANICAL [dry-bulb]
Mechanical. The upper bound on intake air is 25.005 C against a 21.0 C limit, so it fails by 4.005 C. It would take a limit 4.005 C higher, or a bound 4.005 C tighter, to change this hour.

10:00 MECHANICAL [dry-bulb]
Mechanical. The upper bound on intake air is 25.555 C against a 21.0 C limit, so it fails by 4.555 C. It would take a limit 4.555 C higher, or a bound 4.555 C tighter, to change this hour.

11:00 MECHANICAL [dry-bulb]
Mechanical. The upper bound on intake air is 25.280 C against a 21.0 C limit, so it fails by 4.280 C. It would take a limit 4.280 C higher, or a bound 4.280 C tighter, to change this hour.

12:00 MECHANICAL [dry-bulb]
Mechanical. The upper bound on intake air is 24.445 C against a 21.0 C limit, so it fails by 3.445 C. It would take a limit 3.445 C higher, or a bound 3.445 C tighter, to change this hour.

13:00 MECHANICAL [dry-bulb]
Mechanical. The upper bound on intake air is 24.165 C against a 21.0 C limit, so it fails by 3.165 C. It would take a limit 3.165 C higher, or a bound 3.165 C tighter, to change this hour.

14:00 MECHANICAL [dry-bulb]
Mechanical. The upper bound on intake air is 24.165 C against a 21.0 C limit, so it fails by 3.165 C. It would take a limit 3.165 C higher, or a bound 3.165 C tighter, to change this hour.

15:00 MECHANICAL [dry-bulb]
Mechanical. The upper bound on intake air is 22.500 C against a 21.0 C limit, so it fails by 1.500 C. It would take a limit 1.500 C higher, or a bound 1.500 C tighter, to change this hour.

16:00 FREE-COOLING
Free cooling. The 90 %-nominal upper bound on intake air is 20.550 C, which is 0.450 C under the 21.0 C plant limit. That bound is the forecast plus a margin of 1.008 C, and the margin is measured, not chosen: 1.008 C of group-conditional forecast error for this hour of day, plus 0.0000 C for how much the plume could move if the wind direction differs from the one planned for, at the measured spread of wind direction over this lead time.

17:00 FREE-COOLING
Free cooling. The 90 %-nominal upper bound on intake air is 20.280 C, which is 0.720 C under the 21.0 C plant limit. That bound is the forecast plus a margin of 0.830 C, and the margin is measured, not chosen: 0.830 C of group-conditional forecast error for this hour of day, plus 0.0000 C for how much the plume could move if the wind direction differs from the one planned for, at the measured spread of wind direction over this lead time.

18:00 FREE-COOLING
Free cooling. The 90 %-nominal upper bound on intake air is 20.000 C, which is 1.000 C under the 21.0 C plant limit. That bound is the forecast plus a margin of 0.864 C, and the margin is measured, not chosen: 0.864 C of group-conditional forecast error for this hour of day, plus 0.0000 C for how much the plume could move if the wind direction differs from the one planned for, at the measured spread of wind direction over this

lead time.

19:00 FREE-COOLING

Free cooling. The 90 %-nominal upper bound on intake air is 18.055 C, which is 2.945 C under the 21.0 C plant limit. That bound is the forecast plus a margin of 1.076 C, and the margin is measured, not chosen: 1.076 C of group-conditional forecast error for this hour of day, plus 0.0000 C for how much the plume could move if the wind direction differs from the one planned for, at the measured spread of wind direction over this lead time.

20:00 FREE-COOLING

Free cooling. The 90 %-nominal upper bound on intake air is 18.055 C, which is 2.945 C under the 21.0 C plant limit. That bound is the forecast plus a margin of 1.096 C, and the margin is measured, not chosen: 1.096 C of group-conditional forecast error for this hour of day, plus 0.0000 C for how much the plume could move if the wind direction differs from the one planned for, at the measured spread of wind direction over this lead time.

21:00 FREE-COOLING

Free cooling. The 90 %-nominal upper bound on intake air is 17.220 C, which is 3.780 C under the 21.0 C plant limit. That bound is the forecast plus a margin of 0.964 C, and the margin is measured, not chosen: 0.964 C of group-conditional forecast error for this hour of day, plus 0.0000 C for how much the plume could move if the wind direction differs from the one planned for, at the measured spread of wind direction over this lead time.

22:00 FREE-COOLING

Free cooling. The 90 %-nominal upper bound on intake air is 16.945 C, which is 4.055 C under the 21.0 C plant limit. That bound is the forecast plus a margin of 0.801 C, and the margin is measured, not chosen: 0.801 C of group-conditional forecast error for this hour of day, plus 0.0000 C for how much the plume could move if the wind direction differs from the one planned for, at the measured spread of wind direction over this lead time.

23:00 FREE-COOLING

Free cooling. The 90 %-nominal upper bound on intake air is 16.665 C, which is 4.335 C under the 21.0 C plant limit. That bound is the forecast plus a margin of 0.953 C, and the margin is measured, not chosen: 0.953 C of group-conditional forecast error for this hour of day, plus 0.0000 C for how much the plume could move if the wind direction differs from the one planned for, at the measured spread of wind direction over this lead time.

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